

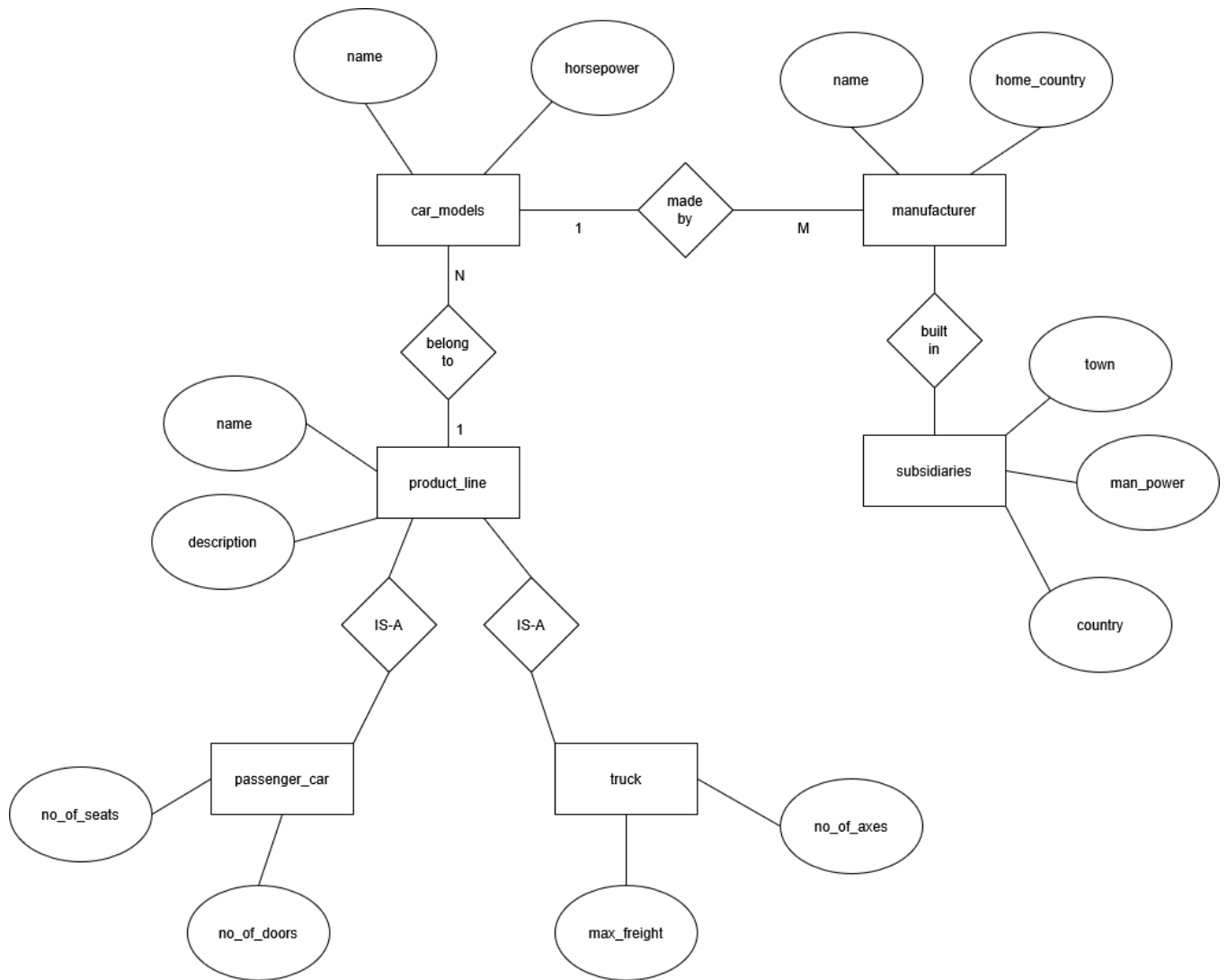
Advanced Database Models Exercise 4

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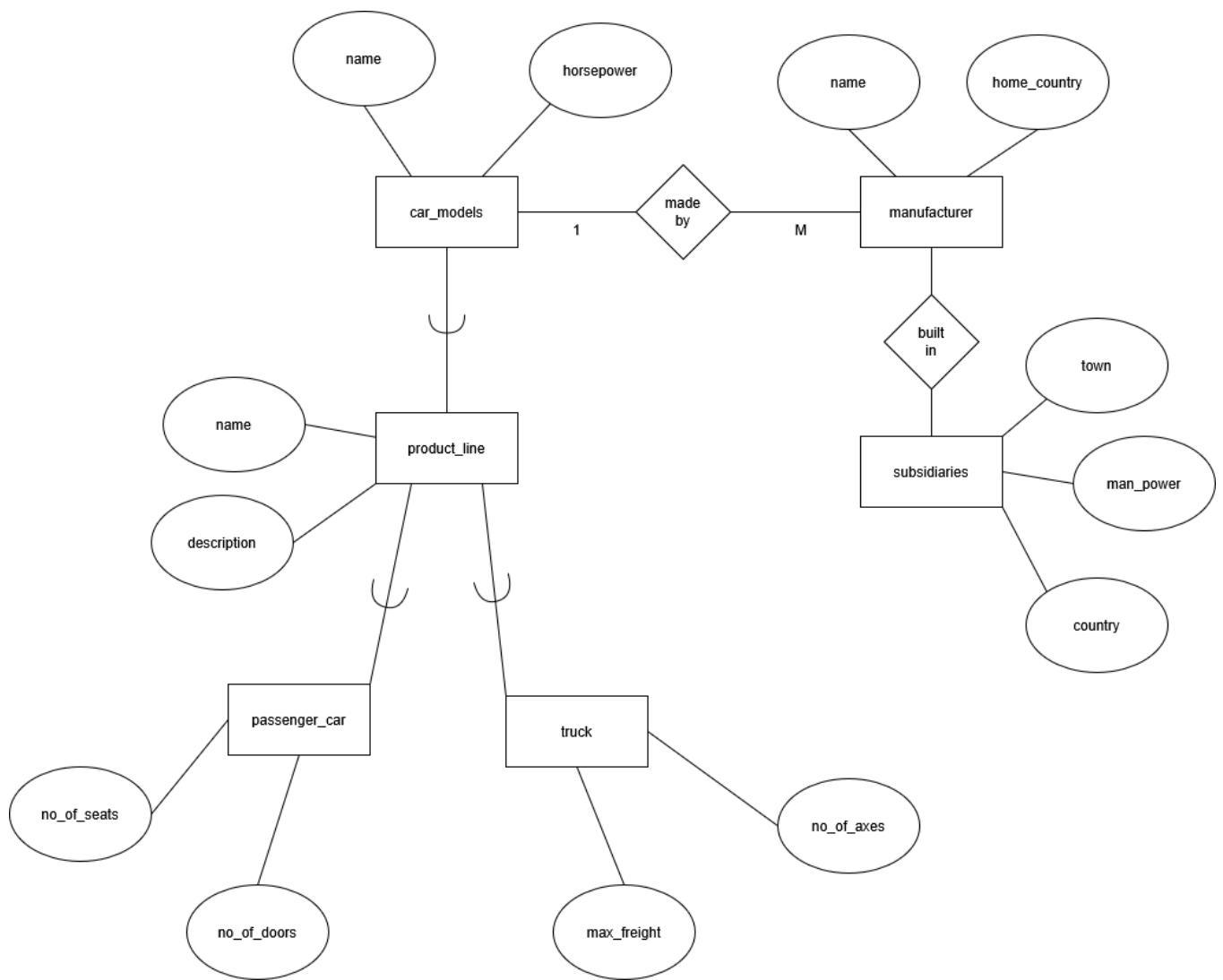
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Question 1 Conceptual Design

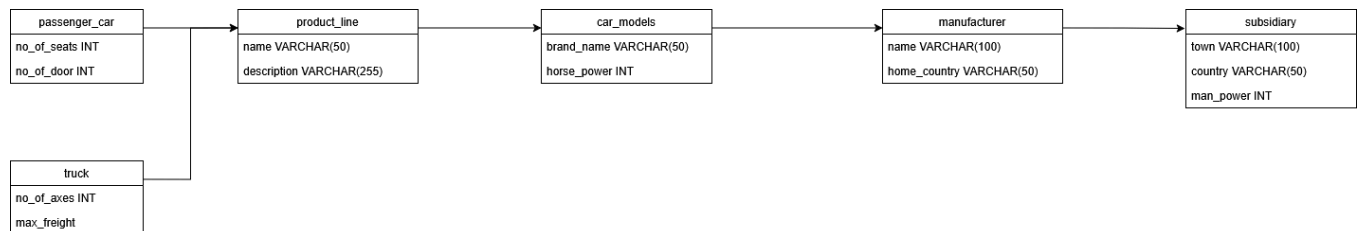
Q1 (a)



Q1 (b)



Q1 (c)



Question 2 Conceptual Design to Logical Design

Q2 (a)

```

artist(ID, name, influenced_by, influences)
singer(s_id, artist_id, sex)
song_writer(sw_id, artist_id, genre)
singer_song_writer(s_id, sw_id)
songs(id, title, duration, s_id, sw_id)
  
```

Q2 (b)

```

import java.util.List;
import java.util.Set;

// Base Class
class Artist {
    private int id;
    private String name;
  }
  
```

```

// Unary/Recursive relationship: An artist can be influenced by many artists,
// and can influence many artists.
private Set<Artist> influencedBy;
private Set<Artist> influences;
}

// Subclass of Artist
class Singer extends Artist {
    private String sex;

    // M:N Relationship represented via the association class/object "Song"
    private List<Song> performedSongs;
}

// Subclass of Artist
class SongWriter extends Artist {
    private String genre;

    // M:N Relationship represented via the association class/object "Song"
    private List<Song> writtenSongs;
}

class SingerSongWriter extends Artist {
    private String sex;
    private String genre;
}

// Association Class for the "performs songs" relationship
class Song {
    private int id;
    private String title;
    private double duration;

    // Links back to the entities involved in the relationship
    private Singer singer;
    private SongWriter songWriter;
}

```